

Curriculum vitae

Erick Arguello Cruz · Pittsburgh, Pennsylvania
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Doctoral research

Carnegie Mellon University

Physics PhD studies · August 2021–present · Advisor: Grigory Tarnopolsky

Theoretical and numerical studies of quantum critical phenomena, including tensor-network simulations, spectral calculations, and finite-size analysis.

- Coauthored a precision study of the massive Schwinger model reporting systems up to 3,000 qubits and agreement across four criticality criteria. [Paper](#).
 - Coauthored work on Yang–Lee quantum criticality and numerical spectra in several dimensions. [Paper](#).
 - Public Julia examples cover lattice-model spectra with ITensors and ITensorMPS; Python notebooks explore random matrices and covariance structure. [Selected computational work](#).
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Education

Carnegie Mellon University

PhD in Physics, in progress · Pittsburgh, PA · August 2021–present

International Centre for Theoretical Physics

Postgraduate Diploma Programme · Trieste, Italy · September 2020–August 2021

Universidad San Francisco de Quito

BS in Physics · August 2013–December 2018

BS in Mathematics · August 2013–December 2017

Teaching and additional research experience

Graduate Teaching Assistant · Carnegie Mellon University
Experimental Physics Laboratory and Quantum Mechanics I.

Full-time Teaching Assistant · Universidad San Francisco de Quito
Quito, Ecuador · January 2018–August 2020

Calculus I and II, Calculus for Sciences I and II, Linear Algebra,
General Physics I, Physics for Sciences I and II, and General Physics
Laboratories I and II.

Research Assistant · Centro Atómico de Bariloche
Magnetic Resonance Laboratory · Bariloche, Argentina · June–July
2018

Operation of a 9 T superconducting magnet and studies of electrical
properties of nanoparticle-decorated graphene.

Research Assistant · Observatorio Astronómico de Quito
Meteorology · Quito, Ecuador · June–July 2016

Meteorological databases and wind prediction models.

Awards

- ICTP Best Performing Diploma Student Award, HECAP section — 2021
- ICTP Diploma Scholarship — 2020
- USFQ Scholarship/Loan — 2018
- USFQ 25 Years Scholarship — 2013

Methods and technical background

Public computational work: Julia, Python, ITensors, ITensorMPS, NumPy, SciPy, pandas, and Matplotlib.

Methods: exact diagonalization, tensor networks, DMRG, spectral statistics, probability, and data analysis.

Additional programming background: MATLAB, Mathematica, and C++.

Working tools: Jupyter, Google Colab, GitHub, Visual Studio, and scientific computing environments.

Languages: Spanish (native), English (fluent).

Additional training

DeepLearning.AI Machine Learning Specialization through Coursera — October 2025–January 2026.

DataCamp studies in supervised learning, deep learning, CNNs, and RNNs — in progress.

Supporting learning exercises.

Interests

Soccer, drawing, literature, boxing, and drums.